12. We define the **group $Z^{(n)}(X)$ of n-cocycles** of X to be the kernel of the coboundary homomorphism $\delta^{(n)}$. We define the **group $B^{(n)}$ of n-coboundaries** of $C^{(n)}(X)$ to be the image of $\delta^{(n-1)}$, that is, it is $\delta^{(n-1)}[C^{(n-1)}(X)]$. Because $\delta^{(n)}[B^{(n)}] = \delta^{(n)}(\delta^{(n-1)}[C^{(n-1)}(X)]) = 0$, we see that $B^{(n)}(X) \leq Z^{(n)}(X)$.

13. The **n-dimensional cohomology group** $H^{(n)}(X)$ of X is $Z^{(n)}(X)/B^{(n)}(X)$.

Computing $H^{(0)}(S)$: We have $B^{(0)}(S) = 0$ because $C^{(-1)}(S) = 0$. Now $\delta^{(0)}(P_1) = P_2P_1 + P_3P_1 + P_4P_1$, and all these summands would have to be "cancelled" for a 0-cocycle. The term P_2P_1 can only be eliminated by another summand P_1P_2, which appears in $\delta^{(0)}(P_2) = P_1P_2 + P_3P_2 + P_4P_2$, and similar observations hold for the summands P_3P_1 and P_4P_1. Thus our only hope for a 0-cocycle is a multiple of $P_1 + P_2 + P_3 + P_4$. Computing, we have

$\delta^{(0)}(P_1 + P_2 + P_3 + P_4) = P_2P_1 + P_3P_1 + P_4P_1 + P_1P_2 + P_3P_2 + P_4P_2 + P_1P_3 + P_2P_3 + P_4P_3 + P_1P_4 + P_2P_4 + P_3P_4 = 0$.

Thus $H^{(0)}(S) \simeq \mathbb{Z}$ and is generated by $(P_1 + P_2 + P_3 + P_4) + \{0\}$.

Computing $H^{(1)}(S)$: If a coset of $B^{(1)}(S)$ in $Z^{(1)}(S)$ contains a 1-cochain c having mP_4P_1 as a summand, then it also contains $c - m\delta^{(0)}(P_1)$, which is a 1-cochain c that does not contain a multiple of P_4P_1. By a similar argument, we can adjust terms of c involving P_2P_4 and P_3P_4 by coboundaries of P_2 and P_3, and we see that the coset contains a 1-cochain of the form $c' = rP_1P_2 + sP_2P_3 + tP_3P_1$. Computing, $\delta^{(1)}(c') = r(P_3P_1P_2 + P_4P_1P_2) + s(P_1P_2P_3 + P_4P_2P_3) + t(P_2P_3P_1 + P_4P_3P_1)$. Now the three 2-simplexes containing the vertex P_4 are all different, so we see that $\delta^{(1)}(c') \neq 0$ unless $r = s = t = 0$, which means $c' = 0$. Thus there are no 1-cocycles that are not 1-coboundaries, so $Z^{(1)}(S) = B^{(1)}(S)$ and $H^{(1)}(S) = 0$.

Computing $H^{(2)}(S)$: Because $C^{(3)}(S) = 0$, every 2-cochain is a cocycle. Examining the coboundaries in $B^{(2)}(S)$, we take the 1-simplex P_1P_2 and find that $\delta^{(1)}(P_1P_2) = P_1P_2P_3 + P_1P_2P_4$. Thus if z is a 2-cycle having $mP_1P_2P_4$ as a summand, we can subtract $m\delta^{(1)}(P_1P_2)$ from Z and obtain another representative of the coset $z + B^{(2)}(S)$ in which $P_1P_2P_4$ does not appear. Now $\delta^{(1)}(P_1P_3) = P_1P_3P_2 + P_1P_3P_4$ and $\delta^{(1)}(P_2P_3) = P_1P_2P_3 + P_2P_3P_4$. Thus by subtracting suitable multiples of these coboundaries, we can find an element of the coset $z + B^{(2)}(S)$ in which no simplex involving P_4 appears. Thus we see that every coset of $B^{(2)}(S)$ contains a unique element of the form $mP_1P_2P_3$ for $m \in \mathbb{Z}$. Clearly $H^{(2)}(S)$ is then infinite cyclic, generated by $P_1P_2P_3 + B^{(2)}(S)$, and isomorphic to $\mathbb{Z}$.

42. Computations of Homology Groups

1. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. Each of the two tangent 1-spheres (circles) is a 1-cycle that is not a 1-boundary, because there is nothing of dimension 2. Thus $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 1$.

2. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. Every 1-cycle is also a 1-boundary. That is, if you cut along the 1-cycle, the surface falls into two disconnected pieces, and the 1-cycle is the 1-boundary of each piece. Thus $H_1(X) = 0$. Each of the two 2-spheres is a 2-cycle, which is not a 2-boundary because there is nothing of dimension 3, so $H_2X \simeq \mathbb{Z} \times \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

3. The space X consists of two disconnected surfaces, the 2-sphere and the annular ring. If P_1 is a vertex in the 2-sphere and P_2 is a vertex in the annular ring, there is no 1-chain having $P_2 - P_1$ as its boundary. Thus $H_0(X)$ is generated by two elements, $P_1 + B_0(X)$ and $P_2 + B_0(X)$; we see that $H_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$. There are no 1-cycles on the 2-sphere that are not 1-boundaries. As we showed

in Example 42.10, any 1-cycle of the annular ring that is not a l-boundary can be "pushed to the outer rim 1-cycle z" in the homology group, so $H_1(X)$ is generated by $z + B_1(X)$. Thus $H_1(X) \simeq \mathbb{Z}$. Finally, there are no 2-boundaries, and the only 2-cycles are multiples of the 2-sphere, so $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

4. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. This time the 1-cycle z which is the outer rim of the annular ring is the 1-boundary of X. There are no 1-cycles that are not 1-boundaries, so $H_1(X) = 0$. The only 2-cycles are multiples of the 2-sphere, so $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

5. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. The only 1-cycles that are not 1-boundaries are multiples of the circle (1-sphere), so $H_1(X) \simeq \mathbb{Z}$. The only 2-cycles are multiples of the 2-sphere, so $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

6. This space is homeomorphic to the torus in Fig. 42.13. (Just let air out of the sphere until it collapses down to be the other half of the "doughnut" surface.) Thus the homology groups are the same as the ones we computed in Example 42.12, namely, $H_0(X) = 0, H_1(X) \simeq \mathbb{Z} \times \mathbb{Z}$, and $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

7. T F F F T F T F F T

8. If P_1 is a vertex on one torus and P_2 is a vertex on the other, then there is no 1-cycle with boundary $P_2 - P_1$, so the 0-cycles are the elements of the cosets $(mP_1 + nP_2) + B_0(X)$ for $m, n \in \mathbb{Z}$. Thus $H_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$. Let a and b be the 1-cycles on the first torus as shown in Fig. 42.13, and let a' and b' be the corresponding 1-cycles on the second torus. The 1-cycles of X that are not 1-boundaries are the elements of cosets

$$(ma + nb + m'a' + m'b') + B_1(X)$$

for $m, n, m', n' \in \mathbb{Z}$ not all zero. Thus we see that $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. Each torus is a 2-cycle, and there are no 2-boundaries, so clearly $H_2(X) \simeq \mathbb{Z} \times \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

9. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. Let b be the 1-cycle which is the circle of intersection of torus 1 with torus 2. Every 1-cycle going that same long way around a torus, like the 1-cycle b in Fig. 42.13, is homologous to our 1-cycle b. (If you cut the surface apart on two circles of this type, the surface will fall into pieces, one of which will have boundary consisting of both circles, so their difference lies in $B_1(X)$. Equivalently, you can "push" one circle, keeping it on the surface, into the other.) We also have two 1-cycles a and a', one on each torus, going the short way around like the circle a in Fig. 42.13. These are not homologous to each other nor to b. Thus the elements of $Z_1(X)$ are the elements of the cosets $(ma + m'a' + nb) + B_1(X)$ for $m, m', n \in \mathbb{Z}$, so $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. Each torus is a 2-cycle and there are no 2-boundaries. We see that $H_2(X) \simeq \mathbb{Z} \times \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

10. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. Let b be the 1-cycle that is the intersection of the torus with the 2-sphere. Now a every 1-cycle that is a circle going the long way around the torus, like the circle b in Fig. 42.13, is homologous to this 1-cycle that is the circle of intersection. (It can be "pushed" into this circle of intersection.) However, if you cut along this circle of intersection, you find that it is the 1-boundary of a hemisphere of S_2, so every 1-cycle of type b is homologous to 0. Thus the only one cycles are those in the cosets $ma + B_1(X)$ for $m \in \mathbb{Z}$, where a is a 1-cycle going the short way around the torus, like the 1-cycle a in Fig. 42.13. We see that $H_1(X) \simeq \mathbb{Z}$. Turning to $H_2(X)$, each